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Theoretical Analysis of Epigenetic Cell Memory by Nucleosome Modification

Ian B. Dodd,[1,2] Mille A. Micheelsen,[1] Kim Sneppen,[1,*] and Geneviève Thon[3]
[1] Center for Models of Life, Niels Bohr Institute, Blegdamsvej 17, DK-2100, Copenhagen Ø, Denmark
[2] Department of Molecular and Biomedical Sciences (Biochemistry), University of Adelaide SA 5005, Australia
[3] Department of Molecular Biology, University of Copenhagen Biocenter, Ole Maaløes Vej 5, DK-2200 Copenhagen N, Denmark
*Correspondence: sneppen@nbi.dk
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SUMMARY

Chromosomal regions can adopt stable and heritable alternative states resulting in bistable gene expression without changes to the DNA sequence. Such epigenetic control is often associated with alternative covalent modifications of histones. The stability and heritability of the states are thought to involve positive feedback where modified nucleosomes recruit enzymes that similarly modify nearby nucleosomes. We developed a simplified stochastic model for dynamic nucleosome modification based on the silent mating-type region of the yeast *Schizosaccharomyces pombe*. We show that the mechanism can give strong bistability that is resistant both to high noise due to random gain or loss of nucleosome modifications and to random partitioning upon DNA replication. However, robust bistability required: (1) cooperativity, the activity of more than one modified nucleosome, in the modification reactions and (2) that nucleosomes occasionally stimulate modification beyond their neighbor nucleosomes, arguing against a simple continuous spreading of nucleosome modification.

INTRODUCTION

Cells carry information handed down from their ancestors and are able to pass on information to their descendants. The majority of this "memory" is encoded in the sequence of bases in the genome of each cell, with nucleic acids providing the high stability and the accurate heritability necessary for memory on evolutionary timescales. Over shorter timescales, cells can also inherit and transmit information that is not stored as changes in their genome sequence. Such "epigenetic" cellular memory involves transient signals setting the cell into one of at least two alternative regulatory states. These states must be *stable* over time and must be *inherited* through cell division. Epigenetic cell memory is particularly important in multicellular organisms, where cells with identical genomes must maintain distinct functional identities, often in similar or identical environments.

One major class of epigenetic mechanisms, sometimes termed "cytoplasmic," is determined by circuits of diffusible, *trans*-acting factors that through positive feedback can exist in alternative regulatory states (Thomas and Kaufman, 2001; Ferrell, 2002; Smits et al., 2006). Well-understood natural examples include stable alternative induction states of the *lac* operon (Vilar et al., 2003), the phage lambda CI-Cro switch (Oppenheim et al., 2005), the *Sxl* sex-determination circuit in *Drosophila* (Bell et al., 1991), and *Xenopus* oocyte maturation (Xiong and Ferrell, 2003). Another class of epigenetic mechanisms, sometimes termed "chromosomal," involves *cis*-specific, DNA-associated differences, the best understood being alternative DNA methylation (Chen and Riggs, 2005). This class of mechanisms is most clearly identified when the alternative states can coexist within the same cell, as seen with stable, heritable inactivation of one of two X chromosomes or monoallelic specific gene expression in mammals (Arney et al., 2001).

A mechanism proposed for epigenetic memory in a number of eukaryotic systems is based on positive feedback loops in nucleosome modification (Grunstein, 1998; Turner, 1998). Nucleosomes package eukaryotic DNA, with a density of about one nucleosome per 200 bp (Felsenfeld and Groudine, 2003). The core nucleosome is composed of two molecules each of four core histones (H2A, H2B, H3, and H4) around which ~150 bp of DNA is wrapped. Nucleosomes may carry various chemical modifications (e.g., acetylation, methylation, or phosphorylation) at different amino acid positions on the different histones, potentially conferring a large information capacity on each nucleosome. Specific additions and removals of these nucleosome modifications are carried out by classes of enzymes, including histone acetyltransferases (HATs), histone methyltransferases (HMTs), histone deacetylases (HDACs), and, more recently discovered, histone demethylases (HDMs; Klose et al., 2006). At least some of these modifications influence the activity of nearby genes, in part because the modifications can affect the binding of regulatory proteins to nucleosomes. Positive feedback can arise if nucleosomes that carry a particular modification recruit (directly or indirectly) enzymes that catalyze similar modification of neighboring nucleosomes. Indeed, some HATs, HDACs, and HMTs are known to associate in vitro or in vivo with histones of the type that they are capable of producing (Jacobson et al., 2000; Owen et al., 2000; Rusché and Rine, 2001; Schotta et al., 2002). Thus, a cluster of nucleosomes may be able to maintain itself stably in a particular modification state. These states are proposed to be inherited through DNA replication because nucleosomes on the parental DNA strand are distributed to both daughter strands (Annunziato, 2005), and the enzymes recruited by these parental nucleosomes may then establish the parental modification pattern on the newly deposited nucleosomes.

To our knowledge there has been no theoretical analysis of the effectiveness of this proposed mechanism and the features required for heritable bistability. In particular, it is known from other systems that positive feedback (or double-negative feedback) is a necessary, but not a sufficient, condition for bistability (Lewis et al., 1977; Ferrell, 2002). Here we examine bistability and heritability using a simplified mathematical model for nucleosome modification-based epigenetic memory, loosely inspired by observations of the silenced mating-type locus of the fission yeast *Schizosaccharomyces pombe* (reviewed in Grewal and Elgin, 2002).

An ~20 kb region of chromosome II of *S. pombe* that contains the two mating-type cassettes, *mat2-P* and *mat3-M*, is normally in a stable "silenced" state, in which the mating-type genes are not expressed. In mutants which have a portion of the silenced region removed and a *ura4*$^+$ reporter gene inserted in the place of the deleted region or nearby (*KΔ::ura4*$^+$), the expression of *ura4*$^+$ and the mating-type genes becomes bistable, flipping between a silenced state where the *ura4*$^+$ gene is repressed and an active state where the *ura4*$^+$ gene is expressed (Grewal and Klar, 1996; Thon and Friis, 1997). Each state is remarkably stable and heritable; transition between them occurs apparently stochastically at roughly equal frequencies of about 5×10^{-4} per cell division (Thon and Friis, 1997; Grewal et al., 1998) or once every 200 days per cell, assuming a 150 min generation time. The memory is chromosomal since two copies of the chromosomal region can exist stably in alternative states within the same cell and can retain their distinct states for many cell generations and through meiosis and sporulation (Grewal and Klar, 1996; Thon and Friis, 1997).

The silenced state of the wild-type locus and the bistable states of the *KΔ::ura4*$^+$ mutants are controlled in a complex manner by histone modifications, HMT and HDAC enzymes, and other factors, including proteins associated with chromatin. The silenced *KΔ::ura4*$^+$ locus and the wild-type locus are highly enriched for nucleosomes in which lysine of histone H3 is methylated (H3K9me) and for the Swi6 protein; these enrichments are absent in active-state *KΔ::ura4*$^+$ cells (Hall et al., 2002). Swi6 belongs to a class of chromodomain proteins capable of binding specifically to H3K9me in vitro (Bannister et al., 2001), and it is, in turn, needed for silencing and H3K9me enrichment at the *KΔ::ura4*$^+$ locus (Hall et al., 2002). Clr4, the *S. pombe* HMT catalyzing the H3K9me modification (Rea et al., 2000), is essential for silencing and for H3K9me and Swi6 enrichment at both the *KΔ::ura4*$^+$ and the wild-type loci (Bannister et al., 2001; Yamada et al., 2005). Clr4 is found in association with silenced centromeric regions (Sadaie et al., 2004) and the wild-type silenced mating-type region (Yamada et al., 2005). It may bind directly to H3K9me via its chromodomain since mutations in its chromodomain affect its ability to methylate histone H3 in vivo but not in vitro, which is indicative of faulty localization (Nakayama et al., 2001), or it might associate with chromatin in a more indirect manner as part of a protein complex (Sadaie et al., 2004; Horn et al., 2005; Thon et al., 2005). The HDAC Clr3 is also crucial to transcriptional silencing. It is needed for the hypoacetylation of H3K14 at the wild-type locus (Bjerling et al., 2002) and for H3K9me and Clr4 enrichment at *KΔ::ura4*$^+$ (Yamada et al., 2005). Like Clr4, Clr3 is physically associated with the wild-type locus, possibly through an interaction with Swi6 (Bjerling et al., 2002; Yamada et al., 2005). Another HDAC, Clr6, acts in parallel with Clr3 to promote silencing (Grewal et al., 1998). Lack of Clr6 is associated with increases in a number of H3 acetylations (Bjerling et al., 2002; Wiren et al., 2005) and a slight increase in H3K9 methylation at the *KΔ* locus (Kim et al., 2004).

The RNAi pathway responsible for pericentromeric gene silencing (Kato et al., 2005; Martienssen et al., 2005; Irvine et al., 2006) is also active at the mating-type locus, but it is unlikely to play a significant role in silencing of the *KΔ::ura4*$^+$ mutants, which lack the *cenH* region (Hall et al., 2002; Jia et al., 2004; Petrie et al., 2005).

Although it is one of the best-characterized epigenetic systems, the *S. pombe* mating-type system is still not sufficiently well understood for detailed model building. Rather, we describe here a highly simplified model that incorporates a number of the features of this system. Despite highly dynamic and noisy behavior of the individual modification events, the model is able to reproduce the strong heritable bistability observed for the *KΔ::ura4*$^+$ mutants. Our analysis reveals fundamental features that are likely to be generally required for nucleosome-based epigenetic systems.

RESULTS AND DISCUSSION

The Model

The basic assumptions in the standard model are as follows:

(1) We consider a DNA region consisting of $N = 60$ nucleosomes. This is equivalent to one nucleosome per 200 bp in the ~12 kb *KΔ::ura4*$^+$ region. The region is isolated from neighboring DNA by boundary elements (Noma et al., 2001; Thon et al., 2002), which we assume to be inert. The modeled region includes the 10.2 kb region stretching between the boundaries and ~1 kb of each

boundary. The number of nucleosomes is assumed to be the same in the silenced and active states.

(2) Only three relevant kinds of nucleosomes are considered: unmodified (U), methylated (M), and acetylated (A). The actual modifications involved are not important for our analysis, and a more general interpretation for U, M, and A could be "unmodified," "modified," and "antimodified." That is, the three distinct nucleosome types may be defined by different kinds of modifications. Although H3K9me modification is a good candidate for the silencing mark in the *S. pombe* system, it is not clear whether H3K9ac or some other mark functions as a mutually exclusive "active mark" in this system. The need for the HDACs Clr3 and Clr6 for silencing provides some support for the idea of acetylation at one or more residues forming an antiH3K9me mark (Grewal and Elgin, 2002). We do not consider different numbers of methyl groups on H3K9, as a role for this is not clear in the *S. pombe* system. Also, although nucleosomes carry two copies of each histone, we are ignoring a role for "hetero-modified" nucleosomes.

(3) Nucleosomes are actively interconverted by modifying and demodifying enzymes (HMTs, HDACs, HDMs, and HATs) that are recruited by the modified nucleosomes as depicted in Figure 1. It is this recruitment that forms the positive feedback in the system. Note that no HDMs or HATs have yet been associated with the mating-type system, and there is thus no evidence for recruitment in these two reactions. However, for simplicity of analysis, we include four symmetrical positive feedback loops in the model (Figure 1) and describe all active recruited processes in terms of the one-rate parameter α.

The term recruitment is used in a general sense to mean that the presence of a modified nucleosome makes it more likely that another nucleosome in the region will be modified. This recruitment might be direct; for example, an HMT might bind directly to an M nucleosome and thus become more likely to methylate a nearby unmodified nucleosome. Recruitment might also be indirect and involve other proteins (e.g., Swi6) or even complex processes such as transcription or DNA replication. No long-term stable association between the modified nucleosome and the recruited enzyme is implied or necessary; in fact the system is expected to be highly dynamic, as observed for many chromatin-associated proteins (Phair et al., 2004).

(4) Nucleosomes can also be interconverted in a recruitment-independent manner. This "noise" in the system can be considered as due primarily to the activity of modifying enzymes that are either free or attached to nucleosomes beyond the region boundaries. Again, in the absence of further information, and for simplicity of analysis, we include four symmetrical random noise interconversions in the model (Figure 1) with the one-rate parameter, $1 - \alpha$. We found that it was the *ratio* of the recruited conversions and the noise conversions that was critical in the system, so an independent noise parameter was not needed.

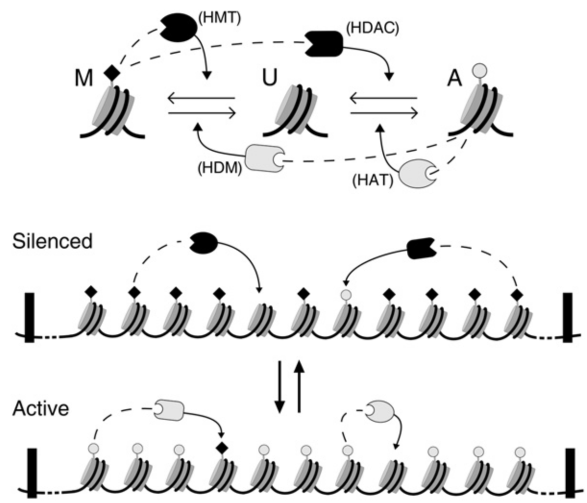

Figure 1. Basic Ingredients of the Model

The three relevant nucleosome types—methylated (M, marked by a black diamond), unmodified (U), or acetylated (A, marked by a gray circle)—can be interconverted by recruitment of histone-modifying enzymes by nearby M or A nucleosomes (dotted lines) or by random "noisy" transitions. HMT indicates histone methyltransferases; HAT indicates histone acetyltransferases; HDM indicates histone demethylases; and HDAC indicates histone deacetylases. Note that within the DNA region delimited by the boundary elements (black rectangles) any nucleosome can stimulate the modification of any other.

(5) We assume that the rates of the interconversion reactions at each nucleosome across the DNA region are the same—that is, homogeneous—with regard to nucleosome position. Deletion of the *K* region and insertion of the *ura4*$^+$ reporter gene in the *K*Δ*::ura4*$^+$ strains changes the balance between silencing and activation, allowing the bistability of the system to become apparent. It is clear that remaining parts of the region—for example silencers near *mat2-P* and *mat3-M* (Ayoub et al., 2000; Thon et al., 1999)—are particularly active in fostering silencing, while other elements (perhaps the *ura4*$^+$ promoter) may foster active chromatin. Although we do not investigate the effects of such heterogeneity here, our preliminary investigations indicate that it does not affect our basic conclusions.

Implementation

The stochastic simulation of the standard model is carried out by iterating the following process of attempted modification of a nucleosome. (Variations of Step 2A are presented later.)

Step 1—A random nucleosome n_1 to be modified is selected among the $N = 60$ nucleosomes. With probability α, a positive feedback (recruited) conversion of n_1 is attempted (Step 2A), OR (with probability $1 - \alpha$), a noisy change of n_1 is attempted (Step 2B).

Step 2A—Recruited conversion: A second random nucleosome n_2 is selected from anywhere within the region, and if n_2 is in either the M or the A state, n_1 is changed one

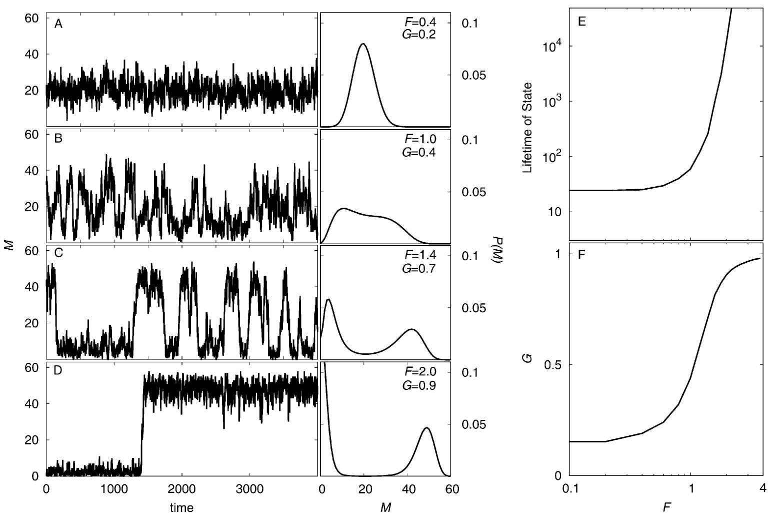

Figure 2. Bistability Is a Function of Noise

(A–D) The left panels show samples of the time development of the number of M nucleosomes, *M,* over a range of feedback-to-noise ratios F = 0.4, 1, 1.4, or 2. The total number of nucleosomes in the system is 60. Time is measured as average attempted conversions per nucleosome. The right panels show the corresponding probability distributions of M obtained from long simulations.

(E) Relationship between F and the average length of time for which the system remains continuously in one or the other state. The high-M state is defined as $M > A$, and the high-A state as $A > M$. Transition to the high-M state is scored when $M > 1.5A$ and to the high-A state when $A > 1.5M$.

(F) Relationship between F and the average "gap" between the numbers of M and A nucleosomes at any time point, $G = \text{Average}(|M - A|/|M + A|)$.

step toward the state of n_2. That is, if n_2 is M, then n_1 is changed A→U or U→M; if n_2 is A, then n_1 is changed M→U or U→A. If n_1 and n_2 are in the same state, or if n_2 is a U, then no changes are performed.

Step 2B—Noisy conversion: Nucleosome n_1 is changed one step toward either of the other types (i.e., no direct A ↔ M interconversions) with a probability of one-third.

The purpose of the one-third probabilities in Step 2B is to make the rates of recruited conversions and noise conversions equal when $\alpha = 1 - \alpha = 0.5$ and the numbers of each type of nucleosome (A, U, and M) are equal such that $A = U = M$.

We found it useful to define a feedback-to-noise ratio, $F = \alpha/(1 - \alpha)$, to convey the relative activities of the positive-feedback and noise-conversion processes (though it should be noted that for a given F, the *actual* ratios of recruited and noise conversions will vary depending on the numbers of the three nucleosome types at the time). Time, t, in the model is defined as average attempted nucleosome conversions per nucleosome. We are aware of no experimental data allowing us to constrain F values or to relate t to real time.

We encourage the reader to consult a Java applet at http://cmol.nbi.dk/models/epigen/Epigen.html (also available as Supplemental Data), which implements the dynamics of the standard model and the variants discussed here (the data for the figures were generated using a C++ program running under Red Hat LINUX).

Bistability of the System Is Controlled by Noise

The left panels in Figures 2A–2D show the dynamics of the system at different feedback-to-noise ratios as F is increased from 0.4 to 2. With decreasing noise, the system displays increasingly two-state behavior, flipping between stable high-M and low-M states. The number of A nucleosomes varies in an opposite manner. This behavior, collected over a long time window, is summarized by the probability distributions shown in the right panels (Figures 2A–2D), which show how often the system contains a given number of M nucleosomes. At low F values, the number of M nucleosomes in the system fluctuates around 20 (one-third of 60), as do the fractions of U and A nucleosomes. As F increases, the distribution broadens, and the system eventually becomes bistable, tending to exist in either the high-M state or the low-M state and tending to avoid states with intermediate levels of M. The number of U nucleosomes (which produces the asymmetry in the probability distributions) gets smaller as F increases.

Figures 2E and 2F show how two different measures of bistability are dependent on the feedback-to-noise ratio, F. Remarkably, the stability of either the high-M state or the low-M state, in terms of the average length of time the system remains in one or the other state, grows faster than exponentially with F (Figure 2E). Thus a sufficiently low noise level can stabilize the states of the system to an arbitrarily high degree. We also quantified bistability by the broadness of the probability distributions of the number of M nucleosomes (Figures 2A–2D, right panels). This is represented by a simple "gap" measure $G = \text{Average}(|M - A| / (M + A))$, which is the absolute difference between the number of M nucleosomes and the number of A nucleosomes (normalized to the maximum possible difference), averaged over a long simulation time. G values close to one signify a close-to-maximal spread of the distribution, indicating strong bistability (Figure 2F). A sharp transition both in G and in the average-state lifetimes occurs as F changes from 0.5 to 2. At $F = 2$—that is, when feedback is only twice as strong as noise—strong bistability is already apparent.

A Need for Cooperativity

It is well known from analysis of regulatory circuits that bistability requires not only positive feedback but also nonlinearity in the feedback loop (Lewis et al., 1977; Thomas and Kaufman, 2001; Ferrell, 2002; Smits et al., 2006). One way in which this can occur is through cooperativity, where, for example, the rate of production of an autoregulatory protein responds to increases in its own concentration in a more-than-linear fashion due to a requirement for multiple copies of the protein in the feedback.

We were thus initially surprised that we could obtain such strong bistability in our model without any need for explicit cooperativity in the recruited conversion process. In the model, each recruitment reaction requires the activity of only one A or M nucleosome in order to cause conversion. Thus the rate at which any U nucleosome is converted to an M nucleosome increases only linearly with M. However, we reasoned that the two-step reaction model (Figure 1) is implicitly cooperative with respect to A↔M conversions since a transition from, for example, an A nucleosome to an M nucleosome, requires two consecutive recruitments by nearby M nucleosomes (one for deacetylation and one for methylation) and thus has a rate proportional to M^2.

To test this possibility, we removed the two-step positive feedback by eliminating either the recruited demodification reactions or the recruited modification reactions. Figure 3 shows that these changes essentially eliminated bistability. High G scores for the one-step recruitment systems were obtained only at very high feedback-to-noise ratios (dashed curves in the left panels of Figures 3B and 3C). The plots in the right panels of Figure 3 show the probability distributions of the systems at very low noise (F = 77). These probability distributions show the fraction of time that the systems spend in states with a particular difference between the number of M and A nucleosomes. (This calculation gives symmetrical distributions, in contrast to the calculation used in Figure 2.) The standard two-step recruitment system spends virtually all of its time equally distributed between the low-M state ($M - A \approx -60$) or the high-M state ($M - A \approx +60$). However, even at this very high F value, the one-step recruitment systems are clearly not bistable (Figures 3B and 3C, right panels, dashed curves).

We asked whether bistability could be restored to the single-feedback systems and whether it could be improved in the standard double-feedback system by adding explicit cooperativity into the recruitment reactions. We did this by making these reactions dependent on *two* other nucleosomes. Specifically, the procedure for recruited conversions was modified to:

Step 2A—Recruited conversion (cooperative): Two nucleosomes n_2 and n_3 are randomly chosen from anywhere within the region, and if these are both in either the M or the A state, then nucleosome n_1 is changed one step toward this state. That is, if n_2 and n_3 are M, then n_1 is changed A→U or U→M; if n_2 and n_3 are A, then n_1 is changed M→U or U→A. If n_2 and n_3 are in different states, or if either is a U, then no changes are performed.

Explicit cooperativity had very little effect on the bistability of the standard two-step feedback system

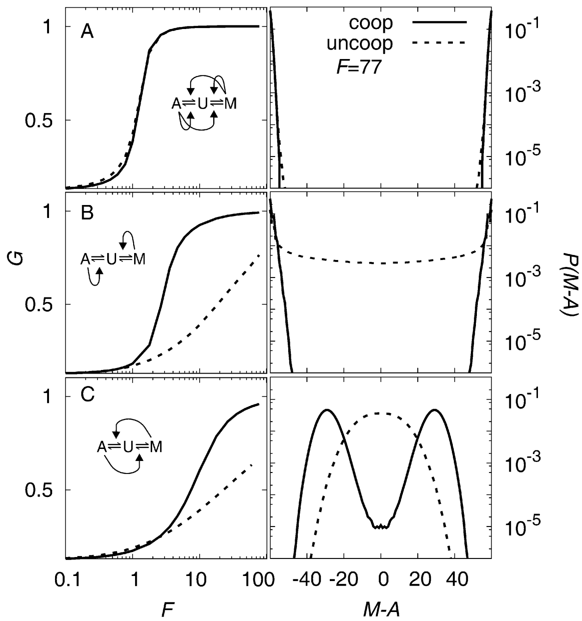

Figure 3. Bistability Requires Implicit or Explicit Cooperativity in the Positive Feedback Loops

The relationship between the gap score G and the feedback-to-noise ratio F is displayed in the left panels, and the probability distributions of the differences between the number of M and A nucleosomes, $M - A$, at F = 77 are displayed in the right panels.

(A) Standard model with recruitment of modifying and demodifying enzymes.

(B) Model with recruitment of modifying enzymes only (U→M and U→A), where Step 2A becomes: A second random nucleosome n_2 from anywhere within the region is selected, and if n_2 is in either the M or the A state and nucleosome n_1 is U, then n_1 is changed to the same state as n_2.

(C) Model with recruitment of demodifying enzymes only, where Step 2A becomes: A second random nucleosome n_2 from anywhere within the region is selected, and if n_2 is in either the M or the A state, respectively, and nucleosome n_1 is A or M, respectively, then n_1 is changed to U. The recruited reactions are either noncooperative (dashed lines), as shown in Figure 1, or are cooperative (solid lines). Cooperativity is introduced by requiring *two* randomly chosen nucleosomes, n_2 and n_3, in the same state (that is, both A or both M) in order for n_1 to be changed (see text).

(Figure 3A, solid curves), presumably because it is already essentially cooperative. In contrast, adding explicit cooperativity substantially improved the bistability of both single-step feedback cases (Figures 3B and 3C, solid curves). Strong bistability could be achieved by the system with cooperative feedback in the modification reactions (U→M and U→A, Figure 3B), although bistabilities equivalent to those obtained by the standard double-feedback model required ~4-fold higher F values. The system with cooperative feedback in the demodification reactions (M→U and A→U, Figure 3C) was only weakly bistable.

Thus, as seen for other positive-feedback systems, our nucleosome-modification model requires explicit or implicit cooperativity to produce bistability.

A Need for "Beyond-Neighbor" Interactions

Models for nucleosome conversion by recruitment that are proposed in the literature generally invoke a linear stepwise process where a modified nucleosome stimulates the modification of its nearest neighbor (Bannister et al., 2001; Rusché and Rine, 2001; Felsenfeld and Groudine, 2003; Grewal and Elgin, 2002). Nucleosome modifications are thus envisioned to spread in a continuous fashion along the DNA. In contrast, our standard model assumes that in the recruitment reaction any nucleosome can act on any other nucleosome in the region, and thus one kind of modification can "jump" over differently modified nucleosomes. "Jumping" might be facilitated by higher-order chromatin structure, by DNA-looping, or by more complex processes such as the passing of an RNA polymerase.

We tested the neighbor-limited contact mechanism by drawing the recruiting nucleosome, n_2, randomly from the two nucleosomes adjacent to n_1. We found that this constraint made bistability much more difficult to achieve than in the standard system. High G scores could be obtained with the neighbor-limited system only at high feedback-to-noise ratios (Figure 4B, left panel). Furthermore, even at these high F values, the neighbor-limited interaction model behaves in a way that seems poorly suited for biological systems. The probability distributions show a large, equiprobable transition region between the two states, with the number of M nucleosomes ranging from about 3 to 57 (Figure 4B, right panel). Within this flat, intermediate zone, boundaries between patches of M or A nucleosomes "wander" along the DNA in random walks, and even a large majority of nucleosomes with one kind of modification is unable to prevent the random growth of patches of nucleosomes carrying the competing modification. The difficulty of obtaining clear two-state behavior in the neighbor-limited interaction model reflects transition dynamics that are similar to those found in the one-dimensional Ising model or to the helix-coil transition in polymer physics (Zimm et al., 1959), reflecting the impossibility of phase transitions in one dimension. One consequence of this behavior is that transitions between the states take longer to execute, an "indecisiveness" that may be undesirable. Another is that the stability of the states is very sensitive to fluctuations in noise; "accidental" introduction of a few nucleosomes of the opposite type (e.g., by nucleosome replacement; Polo et al., 2006) is enough to cause loss of the state. In contrast, the unconstrained system is pushed strongly away from intermediate states (Figure 4A) and thus tends to return to the original state even after large fluctuations; when it does undergo transitions, these are rapid.

We tested whether bistability could be achieved by a nucleosome-nucleosome contact regime intermediate between the unrestricted model and the neighbor-only model. We examined a power-law contact model where contacts between nucleosomes decrease with increasing distance between them on the DNA. Most models for contacts between proteins bound to DNA employ a power law

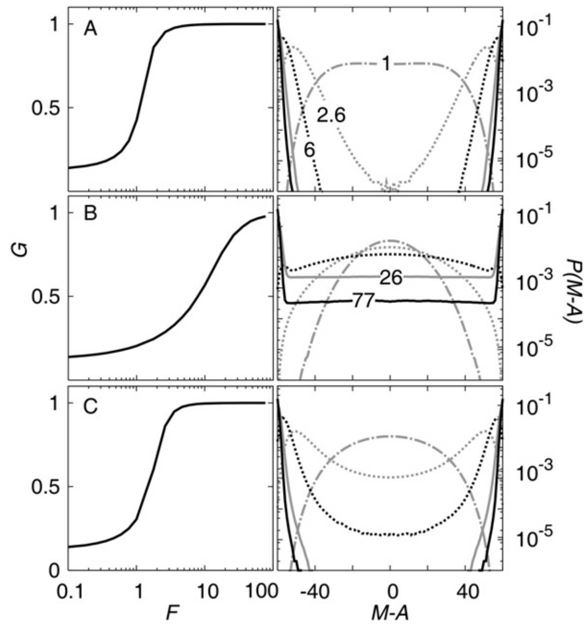

Figure 4. Neighbor-Limited Contacts Produce Poor Bistability
The relationship between the gap score G and the feedback-to-noise ratio F is displayed in the left panels, and the probability distributions of the differences between the number of M and A nucleosomes, $M - A$, at different F values are displayed in the right panels. In the right panels, the F values (1, 2.6, 6, 26, and 77) are indicated on the curves, with the line patterns being the same in (A)–(C) .
(A) Standard model with no spatial constraints in the recruitment reactions.
(B) Neighbor-limited model in which nucleosomes can only stimulate conversion of adjacent nucleosomes (nucleosome n_2 is selected randomly from the two nucleosomes adjacent to n_1).
(C) Power-law contact model in which the probability of one nucleosome stimulating the conversion of another decreases with increasing distance between the two nucleosomes. That is, nucleosome n_2 is selected from nucleosomes d positions away from n_1 with relative probabilities $1/d^{1.5}$ (see text; the probabilities are normalized so that they sum to one).

where the probability of contact is proportional to $1/(d^{1.5})$, where d is the distance between the proteins along the DNA (Ringrose et al., 1999; Rippe, 2001). In Figure 4C, we show the results of this contact scheme, using $1/(d^{1.5})$ as the probability. If d is measured in units of number of nucleosome steps, then for a given nucleosome, the relative probability of it contacting another nucleosome four nucleosomes away ($1/4^{1.5}$) is ~one-eighth of its probability of contacting its nearest neighbor ($1/1^{1.5} = 1$). We found that strong bistability could be obtained at F values that were reasonably low (Figure 4C) but higher than those required in the unrestrained contact regime. It is clear that contact does not have to be completely unconstrained for bistability. The existence of a low rate of longer-range contacts is all that is necessary to allow robust stability of both states. We have not investigated nonpower-law interaction regimes. Although it is known that a simple power law is not adequate to describe very short-range interactions because of the resistance of short DNA segments to bending and twisting (Ringrose et al., 1999; Rippe, 2001), this effect should not be significant for our system, which is dependent on longer-range contacts.

The Effect of DNA Replication

We have so far only examined system bistability. The other critical feature of an epigenetic system is inheritance. We therefore investigated the ability of the high-*M* and low-*M* states to be maintained through DNA replication.

Upon DNA replication, the parental nucleosomes are believed to be partitioned randomly between the two daughter strands, and new nucleosomes are inserted or assembled to fill the gaps (Annunziato, 2005). The modification status of new nucleosomes inserted after replication is not known in *S. pombe*. To keep the model as simple as possible and symmetrical, we assumed initially that all new nucleosomes are in an unmodified form (U)—that is, having no modifications relevant to the M↔A transitions. We accordingly supplemented our standard model with DNA replication and cell division at certain fixed time intervals, with the generation time measured in units of number of attempted nucleosome conversions per nucleosome. At these fixed times, the random partitioning of nucleosomes at replication was simulated by replacing each nucleosome by a U nucleosome with probability of one-half.

The contour plots of Figure 5 show the rate of loss of either the high-*M* (or low-*M*) state per cell generation (or cell cycle) as a function of the feedback-to-noise ratio *F* and generation time, t_{gen}. These data show that high stabilities can be achieved at modest feedback-to-noise ratios despite the destabilizing effect of frequent cell divisions. The stabilities of $\sim 5 \times 10^{-4}$ transitions/cell generation that were observed in the *KΔ::ura4*$^{+}$ strains (Grewal et al., 1998; Thon and Friis, 1997) can be achieved with *F* as low as four and nucleosome-modification rates as low as ten per nucleosome per generation (once every 15 min). In fact, once $F \geq 2$ and once there are more than ~ten modification attempts per nucleosome per cell division, the number of switches per generation is independent of cell-cycle length. This reflects that transitions are much more likely to occur shortly after DNA replication than at any other time point in the cell cycle. Therefore, the model can produce a stability of the modification states that is robust to variations in cell-generation time.

Stable inheritance of the high-*M* or low-*M* states is reduced but not abolished if A and M nucleosomes are among the new nucleosomes inserted after replication. For example, a switching rate of 6.7×10^{-3} per generation with $F = 4$ and $t_{gen} = 30$ is obtained if the new nucleosomes are a mixture of A, U, or M with equal probability; this compares with a switching rate of 1.4×10^{-4} per generation when only U nucleosomes are added (Figure 5). Although the incorporation of the mixed nucleosomes causes a 50-fold destabilization, a significant stability is retained. This is understandable because even after conversion of

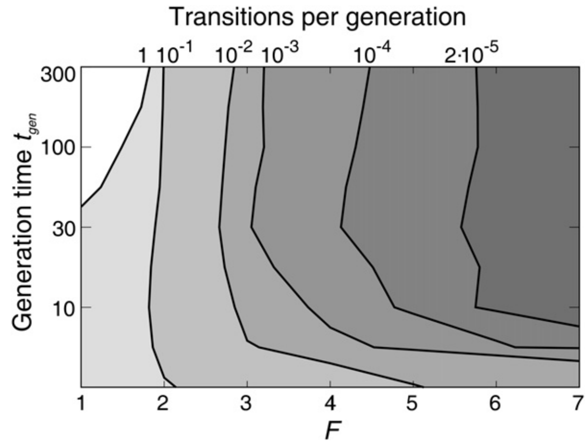

Figure 5. Inheritance of Epigenetic States

Average switching rate per cell cycle as a function of the feedback-to-noise ratio *F* and generation time t_{gen}. DNA replication is simulated by replacing each nucleosome with an unmodified one (U) with a probability of one-half. Generation time is measured as average attempted nucleosome conversions per nucleosome per DNA replication event. Switching is defined as in the legend to Figure 2E.

one-sixth of the nucleosomes to the opposite type, the high-*M* or the low-*M* system remains on the same side of the $M = A$ transition point and experiences a strong "force" to return to the original state (see Figure 4A).

When only unmodified nucleosomes are added after replication, the high-*M* or the low-*M* states of the neighbor-limited contact model can be inherited. However, stabilities are much lower than for the unlimited contact model, even at very high *F* values. For example, a transition rate of 2×10^{-2} per generation is obtained with $F = 77$ and $t_{gen} = 30$. Furthermore, inheritance is essentially abolished if small numbers of A and M nucleosomes are introduced after replication. This is because addition of only a few nucleosomes of the opposite type repositions the system in the large, equipotential transition zone (see Figure 4B), and it is not difficult for the system to wander from there to the other stable state.

Experimental Tests

Current experimental techniques for examining the modification state of nucleosomes at specific locations do not have the ability to resolve changes over short times and within single cells that would be necessary to observe the proposed dynamical interconversions. However, the model makes testable predictions.

Bistability is strongly dependent on the number of nucleosomes in the system, increasing or decreasing exponentially as *N* is increased or decreased. This relationship is shown in Figure 6A, for the standard model with DNA replication. Parameters were chosen ($F = 3.5$ and $t_{gen} = 30$) that produce the observed stabilities of the silenced and active states of the *S. pombe KΔ::ura4*$^{+}$ strains when $N = 60$. Decreasing *N* 2-fold (to 30) produces an ~30-fold

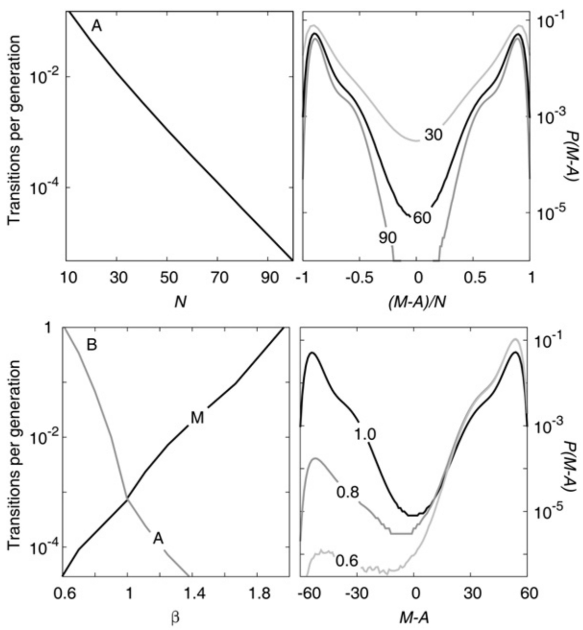

Figure 6. Effect of System Size and Modification Asymmetry on Bistability

Simulations are of the standard model (cooperative, no spatial constraints in the recruitment reactions) with F = 3.5 and t_{gen} = 30.

(A) Varying system size. The left panel shows the number of transitions per generation as a function of system size, N. The right panel gives the probability distributions of the differences between the number of M and A nucleosomes (normalized for N) for N = 30, 60 (the standard system), and 90.

(B) An asymmetric version of the model in which recruitment-stimulated U→A reactions occur with a probability that is reduced by multiplying by a factor β. The left panel shows how the stabilities of the high-A and high-M states change with β (β = 1 is the standard model). The right panel shows probability distributions of the differences between the number of M and A nucleosomes at selected β values (indicated).

reduction in the stabilities of the states; doubling N (to 120) produces an ~650-fold increase in stability (by extrapolation of Figure 6A, left panel). Thus, deletions or insertions of neutral DNA in the *KΔ::ura4*$^+$ region should have predictable and observable effects on the transition rates between silenced and active states.

Bistability is also affected by the relative rates of the recruitment reactions. We found that the introduction of asymmetries in the rates of the reactions in the different directions (M→A versus A→M) produced relatively large differences in the stabilities of the two states. Figure 6B shows the destabilization of the high-A state if the efficiency of the recruited U→A reaction is reduced by a factor β relative to the other three recruited reactions. Increases in reaction efficiencies also have similar strong effects on the relative stabilities of the two states (Figure 6B, left panel). This sensitivity could be demonstrated by measuring the stabilities of the states in an experimental setup in which the expression of a limiting component would be reduced or increased to various levels relative to wild-type (for example by placing its gene under control of the thiamine-regulated nmt1 promoter; Maundrell, 1993).

Conclusions

Our simplified model for epigenetic memory by nucleosome modification provides some unexpected insights.

First, the model can produce high stabilities and heritability of silenced or active states of the DNA region, depending on the feedback-to-noise ratio in nucleosome modification. The stabilities observed for the epigenetic states of the *S. pombe KΔ::ura4*$^+$ strains can be achieved with surprisingly low activities for recruited modification and quite large rates of random conversions. For example, a transition rate as low as 1 per 7000 cell generations can be achieved when there are as few as 8 attempted feedback modifications and 2 noisy modifications per nucleosome per cell cycle (i.e., at F = 4 and t_{gen} = 10). Bistability was also apparent over reasonably broad ranges of the critical parameters, N, F, and t_{gen}, and also with different reaction schemes and nucleosome-contact mechanisms. Thus, our findings support the idea that positive feedback loops in nucleosome modification are an effective and robust mechanism for epigenetic memory.

Second, nucleosome modification can be highly dynamic without compromising stability. A number of authors seem to express a belief that an ideal epigenetic mark should be stable (e.g., Kubicek and Jenuwein, 2004). In contrast, our modeling shows that the state of the whole system can remain stable even when each nucleosome in the system changes its modification status multiple times in a cell generation. This is reminiscent of the observation that a cluster of DNA methylation marks may show an overall stability that is significantly higher than the stability of any single mark (Pfeifer et al., 1990). A benefit of dynamic systems is that they can change state rapidly, a feature that is likely to be important in signal-dependent regulation of promoters controlled by histone modifications.

Third, effective bistability requires cooperativity, either explicit or implicit, in the positive feedback loops in the model. The most effective mechanism for producing bistability was the cooperativity resulting from an ability of modified nucleosomes to not only stimulate addition of the same modification on nearby nucleosomes but also to stimulate removal of competing modifications. Such a mechanism has been suggested to explain the role of HDACs in *S. pombe* silencing (Grewal and Elgin, 2002), and we propose that HDMs play a similar role in stabilizing epigenetic states for which nucleosome methylation is a competing modification. However, bistability can be achieved without direct destabilization of opposing modifications as long as the recruited modification reaction is dependent on the presence of at least *two* modified nucleosomes of the correct type. Such a requirement could be fulfilled if a dimeric histone modifying enzyme needed to bind to two nucleosomes (perhaps by binding to adjacent nucleosomes) in order to catalyze a reaction on a third nucleosome. Thus we predict that systems in which there are only single feedback loops, such as postulated for mating-type cassette silencing in *S. cerevisiae* (Rusché and Rine, 2001), will require this cooperativity. We know of no experiment that addresses explicit cooperativity in nucleosome modification reactions, although Swi6 and the *Drosophila* homolog of Clr4, SU(VAR)3-9, are known to dimerize (Cowieson et al., 2000; Eskeland et al., 2004).

Fourth, we predict that in nucleosome-based epigenetic systems, modified nucleosomes must be able to stimulate the conversion of non-adjacent nucleosomes. If only adjacent nucleosomes can be modified, then a much higher feedback-to-noise ratio is needed to obtain bistability, and the stability of the states is much less resistant to DNA replication and is very sensitive to incorporation of small numbers of anti-modified nucleosomes, for example, after nucleosome exchange (Polo et al., 2006). The requirement for longer range recruited modification is not at variance with the idea that nucleosome modifications in epigenetic systems can spread along the chromosome (e.g., Renauld et al., 1993), but argues against this propagation being mediated solely by contacts between adjacent nucleosomes. We note also that if all types of nucleosome modifications can jump to nonneighboring nucleosomes it is more difficult to construct effective barriers to the spreading of nucleosome modification. In particular, it is challenging to see how a small nucleosome-free region, proposed to be a barrier by Bi et al. (2004), could function in a purely nonlocal model. These difficulties can be remedied, for example, by requiring that at least one type of recruitment process be spatially limited while the others remain nonlocal.

Although the *S. pombe* silent mating-type locus has been the focus of numerous experimental studies over the past ten years and is remarkably well defined from a biological point of view, the system is at a point where it could benefit from a mathematical model to provide a framework for fundamental experiments. We also expect that the model can be extended to other systems and hope that it will stimulate a broader audience of scientists interested in epigenetic effects.

Supplemental Data

Supplemental Data include a Java applet and can be found with this article online at http://www.cell.com/cgi/content/full/129/4/813/DC1/.

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